$$v_3' = ev_3 \tag{17}$$

$$E_{k3}' = e^3 E_{k3} \tag{18}$$

$$E_{k4} = E_{k3}' + qU \tag{19}$$

（3）、（15）、（18）、（19）

$$E_{k4} = \left(1 + e^2 + e^4\right) C_0 U^2 \tag{20}$$

$$v_4 = \sqrt{\frac{2E_{k4}}{m}} \tag{21}$$

$$v_4' = ev_4 \tag{22}$$

$$E_{k4}' = e^2 E_{k4} \tag{23}$$

n

$$E_{kn} = \left[1 + e^2 + \cdots + e^{2(n-2)}\right] C_0 U^2 \tag{24}$$

$$E_{kn}' = e^2 \left[1 + e^2 \cdots + e^{2(n-2)}\right] C_0 U^{22} \tag{25}$$

$$E_{kn} = \frac{1 - e^{2(n-1)}}{1 - e^2} C_0 U^2 \tag{26}$$

$$E_{ln}' = \frac{e^2 \left[1 - e^{2(n-1)}\right]}{1 - e^2} C_0 U^2 \tag{27}$$

$e < 1$ E_{kn} E_{kn}' $n \to \infty$

$$E_{k?} = \frac{1}{1 - e^2} C_0 U^2 \tag{28}$$

$$E_{k\infty}' = \frac{e^2}{1 - e^2} C_0 U^2 \tag{29}$$

$$E_{kmax} = E_{k\infty} = \frac{1}{1 - e^2} C_0 U^2 \tag{30}$$

2. $n \to \infty$

$$v_\infty' = \sqrt{\frac{2E_{k\infty}'}{m}} = eU \sqrt{\frac{2C_0}{\left(1 - e^2\right) m}} \tag{31}$$